

CR30 / ELECTRONIC CAM FIELD AUDIT

KOSME TOP II AD TANDEM 20/1056 CE SA4 E2 | Registration L05358 | Manual K966-749

PURPOSE	Capture the existing condition, identify one-board versus all-board communication failure, verify every address and signal path, and document any controlled repair without losing the original state.
DO NOT SKIP	Photograph and record before touching. Make one change at a time. Do not clone again until the source card, target address, network power and communication chain are verified.

1. Job and Machine Identification

Audit date / time		Technician(s)	
Work order / event ID		Production line / location	
Product / bottle format		Current recipe / cam number	
Machine state on arrival	☐ Running ☐ Stopped ☐ Intermittent	Production impact	☐ Down ☐ Reduced ☐ Quality issue
Control/display identity	CR30 (page heading) / KET39 (manual index alias)	Platform/head count	20 documented; observed: _____

2. Safety and Evidence Gate

☐ Exact error photographed before reset	☐ CR30 screen and full panel photographed
☐ Every card, connector and DIP bank photographed before change	☐ Plant LOTO applied before guards/covers were removed
☐ OEM minimum one-minute wait observed after power isolation	☐ Required PPE / energized-work controls applied for any justified live measurement
☐ Guards and safety switches restored before powered test	☐ One-change-at-a-time rule accepted

DIP notation rule: record the physical ON arrow/mark, bank orientation, switch count, and exact left-to-right pattern. Do not assume a binary convention until the board marking or schematic confirms it.

3. Initial Fault Snapshot - Complete Before Resetting

Exact CR30/HMI message	
Error number(s)	☐ 24 Software ☐ 25 Hardware ☐ 26 High temperature ☐ 27 COM ☐ 28 Faulty line Other: _____
When fault occurs	☐ Power-up ☐ RUN before zero ☐ After zero ☐ During production ☐ After format change ☐ After clone
Scope	☐ One board/card ☐ Several boards ☐ All boards ☐ Unknown
Board/head identified	
Beacon / machine state	☐ Red-stop ☐ Yellow-warning ☐ Green-running ☐ Blue-reset ☐ RUN ☐ READY
Can error be reset?	☐ Yes and stays clear ☐ Yes but returns ☐ No Time to return: _____
Observed symptom	☐ No label ☐ Misplaced label ☐ Plate mismatch ☐ No centering ☐ No communication ☐ Other: _____

4. Replacement / Clone History

Replacement card installed?	☐ No ☐ Yes	Date / technician	
Replacement card board/head		DIP/address set to	
Source card used for clone		Was source different from target?	☐ Yes ☐ No ☐ Unknown
Machine code entered		OEM candidate	05358 - confirm on actual display
Head quantity entered		Expected candidate	20 - verify before reuse
Copy counter reached 0	☐ Yes ☐ No ☐ Unknown	Power interrupted?	☐ No ☐ Yes ☐ Unknown
Startup result	☐ RUN seen ☐ Machine zero achieved ☐ READY seen	Failure point	

Exact sequence already performed and what happened

5. Electronic Cam Card Inventory - Cards / Heads 1-10

Record values before changing DIP switches, connectors or programs. DIP pattern field is intentionally unrestricted because the switch-bank size and ON orientation must come from the actual board.

Card / head	Photo ID	DIP pattern as found	Decoded address	Board power V	Communicate s?	Ve firmware	Tp temp C	Pc changes?	En position	Sx / notes
1					[]Y []N			[]Y []N		
2					[]Y []N			[]Y []N		
3					[]Y []N			[]Y []N		
4					[]Y []N			[]Y []N		
5					[]Y []N			[]Y []N		
6					[]Y []N			[]Y []N		
7					[]Y []N			[]Y []N		
8					[]Y []N			[]Y []N		
9					[]Y []N			[]Y []N		
10					[]Y []N			[]Y []N		

Address conflicts / firmware mismatches / abnormal temperatures noted:

6. Electronic Cam Card Inventory - Cards / Heads 11-20

Record values before changing DIP switches, connectors or programs. DIP pattern field is intentionally unrestricted because the switch-bank size and ON orientation must come from the actual board.

Card / head	Photo ID	DIP pattern as found	Decoded address	Board power V	Communicate s?	Ve firmware	Tp temp C	Pc changes?	En position	Sx / notes
11					[]Y []N			[]Y []N		
12					[]Y []N			[]Y []N		
13					[]Y []N			[]Y []N		
14					[]Y []N			[]Y []N		
15					[]Y []N			[]Y []N		
16					[]Y []N			[]Y []N		
17					[]Y []N			[]Y []N		
18					[]Y []N			[]Y []N		
19					[]Y []N			[]Y []N		
20					[]Y []N			[]Y []N		

Address conflicts / firmware mismatches / abnormal temperatures noted:

7. Common Power, Safety and Communication Audit

Check	As found	Result / measurement	Photo / evidence ID	Changed?
Board-supply circuit breaker condition	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
CR30 supply: terminal 1 +24 V / terminal 2 common	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Board power present at first, middle and last card	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Head safety switches fully closed	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Plates coupled / applicable centering photocells on	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
CR30-to-head connectors fully seated	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Sliding contacts clean, aligned and making contact	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Flat signal cable free of tears, folds and pin damage	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Flat-cable connectors seated and strain relieved	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Power connectors seated; cable screws tight	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Shared common/negative continuity verified	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Duplicate DIP/network addresses ruled out	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Card firmware versions compatible / compared	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Metal particles / contamination inside boards checked	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Board-to-motor cable insulation checked (Error 25)	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes
Ventilation / card temperature checked (Error 26)	☐ OK ☐ Fault ☐ N/A			☐ No ☐ Yes

8. Error 27 Branch Test

ONE BOARD REPORTS COM ERROR	ALL BOARDS REPORT COM ERROR
1. Verify power at that board. 2. Inspect power cable and connector. 3. Tighten applicable cable screws. 4. Inspect that board's flat-cable connection. 5. Confirm unique DIP address. 6. Compare firmware and card data. 7. Replace only after these checks remain good.	1. Verify common board power and breaker. 2. Verify head safety switches. 3. Inspect CR30-head connections and sliding contacts. 4. Inspect flat cable. 5. Connect one board only and test. 6. Add boards individually until failure returns. 7. Investigate the board/cable segment that triggers failure.

Board-by-board isolation log - begin with one known/likely good board. Record the physical cable segment changed at each step.

Step	Board added / removed	Address	Power V	CR30 communication	Error returned?	Cable / connector segment	Time	Notes
1				☐ OK ☐ Fail	☐ Y ☐ N			
2				☐ OK ☐ Fail	☐ Y ☐ N			
3				☐ OK ☐ Fail	☐ Y ☐ N			
4				☐ OK ☐ Fail	☐ Y ☐ N			
5				☐ OK ☐ Fail	☐ Y ☐ N			
6				☐ OK ☐ Fail	☐ Y ☐ N			
7				☐ OK ☐ Fail	☐ Y ☐ N			
8				☐ OK ☐ Fail	☐ Y ☐ N			
9				☐ OK ☐ Fail	☐ Y ☐ N			
10				☐ OK ☐ Fail	☐ Y ☐ N			
11				☐ OK ☐ Fail	☐ Y ☐ N			
12				☐ OK ☐ Fail	☐ Y ☐ N			

Isolation conclusion - first step that caused communication failure

9. Controlled Replacement and Program-Copy Verification

Complete only after power, addresses and communication are stable. A network-wide copy can overwrite every online card, so the source must be proven good.

☐ Machine isolated; replacement installed; connectors verified	☐ Replacement card DIP address documented and unique
☐ Source card selected is known-good and is NOT the replacement card	☐ Source card address: _____ Target/replacement address: _____
☐ Language selection recorded: _____	☐ Head quantity entered: _____ (documented candidate: 20)
☐ Machine code entered: _____ (OEM candidate: 05358; confirm)	☐ Copy started without communication error
☐ Decrementing copy counter reached zero	☐ No power interruption / connector disturbance during copy
☐ CR30 reset and displayed RUN	☐ First machine zero was achieved
☐ CR30 displayed READY	☐ Card firmware and parameter readout compared after copy
☐ Production test completed at controlled speed	

Copy result / exact message / failed step

Post-copy functional test: products run, heads affected, label result

10. CR27 Synchronizer and Machine-Zero Audit

CR27 supply	+12 V observed: _____ Common observed: _____	Display active?	☐ Yes ☐ No
Operating display	Fo active size: _____	Next plate advances?	☐ Yes ☐ No
th head/platform count		Expected candidate	20 - verify actual value
Pt pulses / revolution		Fo format	
Zero sensor sequence	☐ Last plate start-photo ☐ Zero sensor ☐ Plate 1 start-photo	Plate 1 correctly established?	☐ Yes ☐ No
Loss-of-zero symptom	☐ Not present ☐ Suspected ☐ Confirmed	Average delay fallback suspected?	☐ Yes ☐ No
Inputs / output	T3 encoder: ____ T4 start photo: ____ T5 zero: ____ T8 group start: ____	Shared common verified?	☐ Yes ☐ No

Plate-specific Pi delay record - record before changing. Increasing delay is described by the OEM as starting the label in advance.

Plate	Pi as found	Label position result	Changed to	Reason / outcome
1		☐ Good ☐ Early ☐ Late		
2		☐ Good ☐ Early ☐ Late		
3		☐ Good ☐ Early ☐ Late		
4		☐ Good ☐ Early ☐ Late		
5		☐ Good ☐ Early ☐ Late		
6		☐ Good ☐ Early ☐ Late		
7		☐ Good ☐ Early ☐ Late		
8		☐ Good ☐ Early ☐ Late		
9		☐ Good ☐ Early ☐ Late		
10		☐ Good ☐ Early ☐ Late		

10A. CR27 Plate-Specific Pi Record - Plates 11-20

Plate	Pi as found	Label position result	Changed to	Reason / outcome
11		☐ Good ☐ Early ☐ Late		
12		☐ Good ☐ Early ☐ Late		
13		☐ Good ☐ Early ☐ Late		
14		☐ Good ☐ Early ☐ Late		
15		☐ Good ☐ Early ☐ Late		
16		☐ Good ☐ Early ☐ Late		
17		☐ Good ☐ Early ☐ Late		
18		☐ Good ☐ Early ☐ Late		
19		☐ Good ☐ Early ☐ Late		
20		☐ Good ☐ Early ☐ Late		

CR27 / machine-zero observations and conclusion

11. CR30 Centering and Platform Compensation Audit

Cam number / name		R1 common lag	
Fu mode	☐ Manual ☐ Auto	Me reference	☐ Centre ☐ Edge
Mp multiplier	☐ SL x1 ☐ SL x4	Sp polarity	☐ Dark ☐ Light
SL minimum spot length		Sr measured spot	
Sg threshold readout		Empty-plate test NET1	☐ Off ☐ On ☐ Unknown

Per-platform Cp compensation - OEM range -11.2 to +11.2 degrees. Final centering position = R1 + Cp. Record original value before adjustment.

Platform	Cp as found	Observed centering	Changed to	Outcome / bottle-photo ID
1		☐ Good ☐ CW ☐ CCW		
2		☐ Good ☐ CW ☐ CCW		
3		☐ Good ☐ CW ☐ CCW		
4		☐ Good ☐ CW ☐ CCW		
5		☐ Good ☐ CW ☐ CCW		
6		☐ Good ☐ CW ☐ CCW		
7		☐ Good ☐ CW ☐ CCW		
8		☐ Good ☐ CW ☐ CCW		
9		☐ Good ☐ CW ☐ CCW		
10		☐ Good ☐ CW ☐ CCW		

11A. CR30 Platform Cp Record - Platforms 11-20

Platform	Cp as found	Observed centering	Changed to	Outcome / bottle-photo ID
11		☐ Good ☐ CW ☐ CCW		
12		☐ Good ☐ CW ☐ CCW		
13		☐ Good ☐ CW ☐ CCW		
14		☐ Good ☐ CW ☐ CCW		
15		☐ Good ☐ CW ☐ CCW		
16		☐ Good ☐ CW ☐ CCW		
17		☐ Good ☐ CW ☐ CCW		
18		☐ Good ☐ CW ☐ CCW		
19		☐ Good ☐ CW ☐ CCW		
20		☐ Good ☐ CW ☐ CCW		

Centering observations, photos and conclusion

12. Findings, Corrective Action and Verification

Evidence-backed failure statement - what failed, when, and on which board(s)			
Primary cause category		☐ Power ☐ Breaker/safety ☐ Address/DIP ☐ Connector ☐ Flat cable ☐ Sliding contact ☐ Board ☐ Firmware/program ☐ Zero/sync ☐ Centering	
Failed component / location			
Evidence proving cause			
Corrective action			
Parts used / part numbers			
Values changed		Original: _____ New: _____	
Production verification		Product/format: _____ Speed: _____ Duration/count: _____ Result: _____	
Recurrence check		☐ Same shift ☐ 24 hours ☐ 7 days ☐ 30 days Owner: _____	
Unresolved findings / escalation required			
Asset Graph event ID		Work order / service ticket	
Photos attached		OEM pages used	65-80
Technician signature		Date / time closed	
Reviewed by		Review date	

Asset Graph handoff: store each card as a separate addressable component; attach DIP pattern, firmware, temperature, communication state, photos and test results. Store every replacement or clone as an event with source address, target address, code, counter result and RUN-to-READY outcome.